ZOO405 – ECOLOGY 100% GUESS

Mid-Term File 2023 By

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Decomposers and their 5 examples:

Decomposers: An organism that gets energy by breaking down the Remains of dead organisms and the wastes of living things. These are the Another kind of heterotroph. They are often called nature's recyclers. They return carbon and nitrogen to the air and soil.

Examples: Most fungi and bacteria, Earthworms and insects , mites , snails, worm, mushrooms.

What do you mean by precipitation?

Precipitation is rain, snow, sleet, or hail — any kind of weather condition where something's falling from the sky. Precipitation has to do with things falling down, and not just from the sky. It's also what happens in chemical reactions when a solid settles to the bottom of a solution.

What is evaporation:

Evaporation is the process that changes liquid water to gaseous water (water vapor). Water moves from the Earth's surface to the atmosphere via evaporation. Evaporation occurs when energy (heat) forces the bonds that hold water molecules together to break.

Differentiate the food chain and food web:

Food chain: Food chain is a linear sequence of organisms through which nutrients and energy pass as one organism eats another. In a food chain, each organism occupies a different trophic level.

Food web: Food webs consist of many interconnected food chains and are more realistic representation of consumption relationships in ecosystems. Energy transfer between trophic levels is inefficient. This inefficiency limits the length of food chains.

Economical Importance of Grasslands:

1= Grasslands are the grazing areas of many rural communities. 2= Farmers who keep cattle or goats, as well as shepherds who keep sheep, are highly dependent on grasslands. 3= The grass is also used to thatch houses and farm sheds.4= Overgrazing by huge herds of domestic livestock has degraded many grasslands. 5= Grasslands have diverse species of insects that pollinate crops.

What are the autotrophs and their types:

The autotrophs are the organisms that can produce their own food using light, water, carbon dioxide, or other chemicals. Because autotrophs produce their own food, they are sometimes called producers. They are the ("self-feeders").

Types:

Two basic types of autotrophs: 1. Photoautotrophs 2. Chemoautotrophs

Photoautotrophs: Such as plants, use energy from sunlight to make organic compounds (sugars) out of carbon dioxide in photosynthesis. Other examples of photoautotrophs include algae and cyanobacteria.

Chemoautotrophs: Chemoautotrophs use energy from chemicals to build organic compounds out of carbon dioxide (or similar molecules). This is called chemosynthesis. For instance, there are hydrogen sulfide-oxidizing chemoautotrophic bacteria found in undersea vent communities (where no light can reach).

What are the Biotic factors and Abiotic factors:

Biotic factors : Biotic factors are defined as the living components or factors that affect an ecosystem or other organisms living in that ecosystem. Bacteria that live inside the intestine of an animal act as biotic factors that have acquired the role of helping with the proper digestion of the food in the intestine.

Example: Biotic factors include animals, plants, fungi, bacteria, and protists.

Abiotic Factors: Abiotic components are the physical and/or the chemical factors that act on the living organisms at any part of their life.

Examples: Sunlight, air, precipitation, minerals, and soil are some examples of abiotic factors.

Nitrogen Cycle:

Nitrogen Cycle is a biogeochemical process through which nitrogen is converted into many forms, consecutively passing from the atmosphere to the soil to organism and back into the atmosphere.

Process of the Nitrogen Cycle consists of the following steps: 1= Nitrogen fixation, 2= Nitrification 3= Assimilation 4= Ammonification 5= Denitrification

1= Nitrogen fixation: It is the initial step of the nitrogen cycle. Here, Atmospheric nitrogen (N2) which is primarily available in an inert form, is converted into the usable form -ammonia (NH3).

2= Nitrification: In this process, the ammonia is converted into nitrate by the presence of bacteria in the soil. Nitrites are formed by the oxidation of ammonia with the help of Nitrosomonas bacteria species.

3= Assimilation: Primary producers – plants take in the nitrogen compounds from the soil with the help of their roots, which are available in the form of ammonia, nitrite ions, nitrate ions or ammonium ions and are used in the formation of the plant and animal proteins. This way, it enters the food web when the primary consumers eat the plants.

4= Ammonification: When plants or animals die, the nitrogen present in the organic matter is released back into the soil. The decomposers, namely bacteria or fungi present in the soil, convert the organic matter back into ammonium.

5= Denitrification: Denitrification is the process in which the nitrogen compounds make their way back into the atmosphere by converting nitrate (NO3-) into gaseous nitrogen (N). This process of the nitrogen cycle is the final stage and occurs in the absence of oxygen.

Define producers:

Plants are producers. They make their own food, which creates energy for them to grow, reproduce and survive. Being able to make their own food makes them unique.

Define ecology:

Ecology is the study of organisms and how they interact with the environment around them. An ecologist studies the relationship between living things and their habitats.

Define Sublimation:

Sublimation occurs when snow or ice changes directly into water vapour without becoming water. It usually occurs as a result of dry winds and low humidity.

what are the autotrophs and hetrotrophs

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Heterotrophs: A heterotroph is an organism that eats other plants or animals for energy and nutrients. The term stems from the Greek words hetero for "other" and trophe for "nourishment."

Importance of the desert ecosystem:

1= Like other ecosystems, desert ecosystems occupy a significant place in the world. This ecosystem plays an important role in maintaining the balance of the earth despite the low rainfall and dry climate. 2= This ecosystem is a habitat for many species of plants and animals. These animals and plants balance by establishing relationships with each other. 3= Different types of bacteria are present in the sand of the desert ecosystem. These bacteria can store carbon dioxide. And acts as a carbon sink, preventing carbon dioxide from entering the atmosphere. 4= These ecosystems are often the source of mineral resources, natural gas.

Diffirentiate between Igneous rocks and sedimentry rocks:

Igneous rocks: igneous rocks are formed by the cooling of hot molten rock also known as magma. When the hot magma cools it begins to harden meaning once it had fully cooled it create what is known to be an igneous rock.

Sedimentary rocks: sedimentary rocks are formed from pre-existing rocks. When rocks erode and mix with other dirt, clay and particles then settle together the mix together to form a sedimentary rock.

Importance of Ecology:

Conservation of Environment: Ecology helps us to understand how our actions affect the environment. It shows the individuals the extent of damage we cause to the environment.

Resource Allocation: With the knowledge of ecology, we are able to know which resources are necessary for the survival of different organisms. Lack of ecological knowledge has led to scarcity and deprivation of these resources, leading to competition.

Eco-Friendliness: Ecology encourages harmonious living within the species and the adoption of a lifestyle that protects the ecology of life.

Energy Conservation: All organisms require energy for their growth and development. Lack of ecological understanding leads to the over-exploitation of energy resources such as light, nutrition and radiation, leading to its depletion.